



A Machine Learning Approach to Patient Perceptions of AI in Clinical Encounters

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Background

- In 2023-2024, the American Medical Association found that physician AI usage surged from 38% to 66% (Henry, 2025).
- While 65% of US hospitals reported use of AI in EHRs, 56% failed to report AI bias evaluations (Nong et al., 2025).
- Ambient AI usage has been linked to decreased cognitive load, with the reasoning of improving interactions with patients instead of focusing on the burden of documentation (Stults et al., 2025).
- Comfort, patient-clinician trust, perceived risk and benefits, and use responsibility were found to be primary drivers of informed consent (Lawrence et al., 2025).
- A sentiment analysis with Chinese social media found that healthcare professionals prioritized sentiments related to the profession, while non-professionals focused on personal impact (Lu et al., 2025).

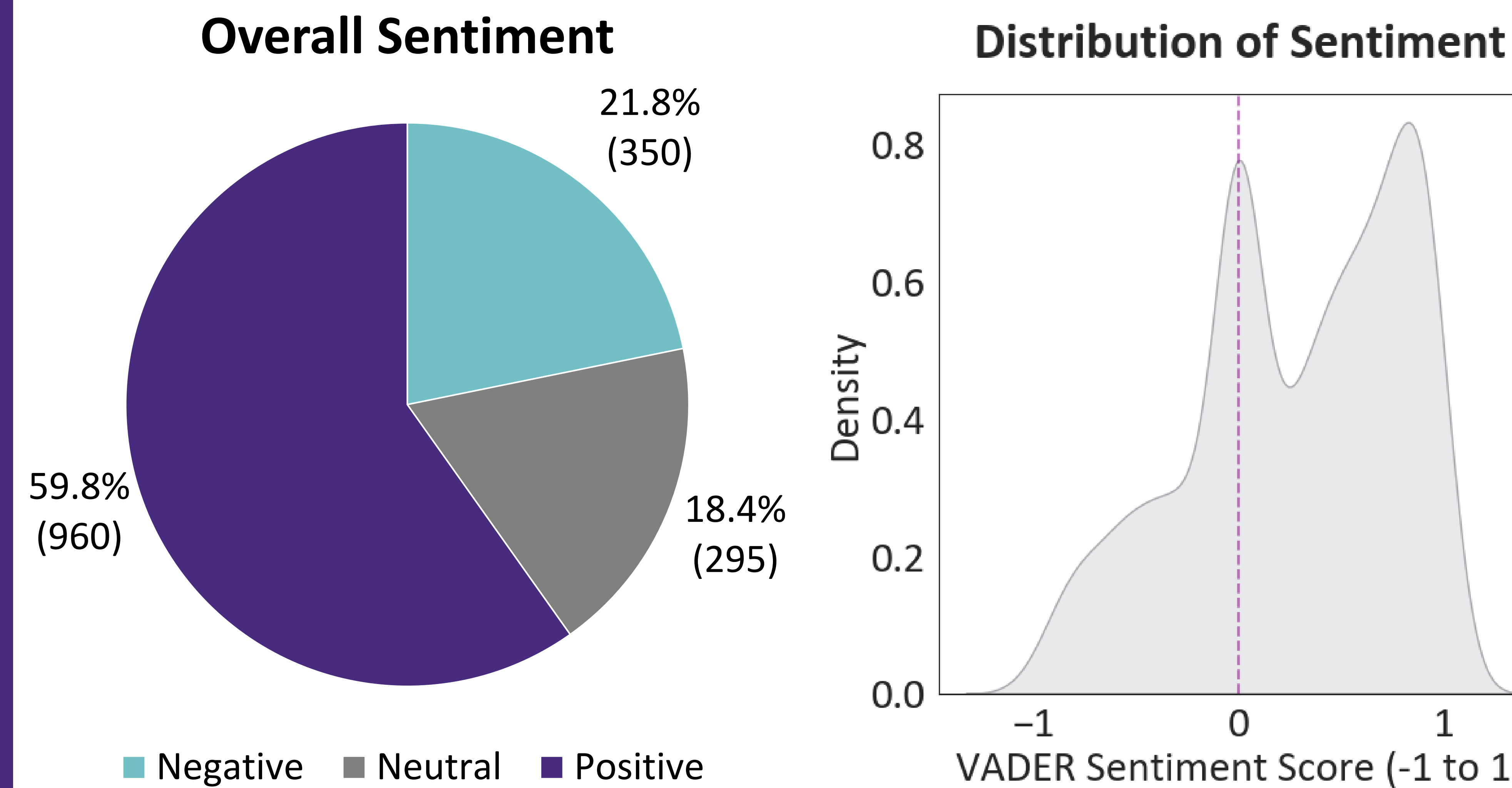
Current Study

- What is the prevailing sentiment among social media (Reddit) users regarding concerns of the use of AI in healthcare settings?
- What specific factors (e.g., data privacy, transparency, perceived benefits/risks, clinician trust) are most frequently associated with these sentiments?

Method

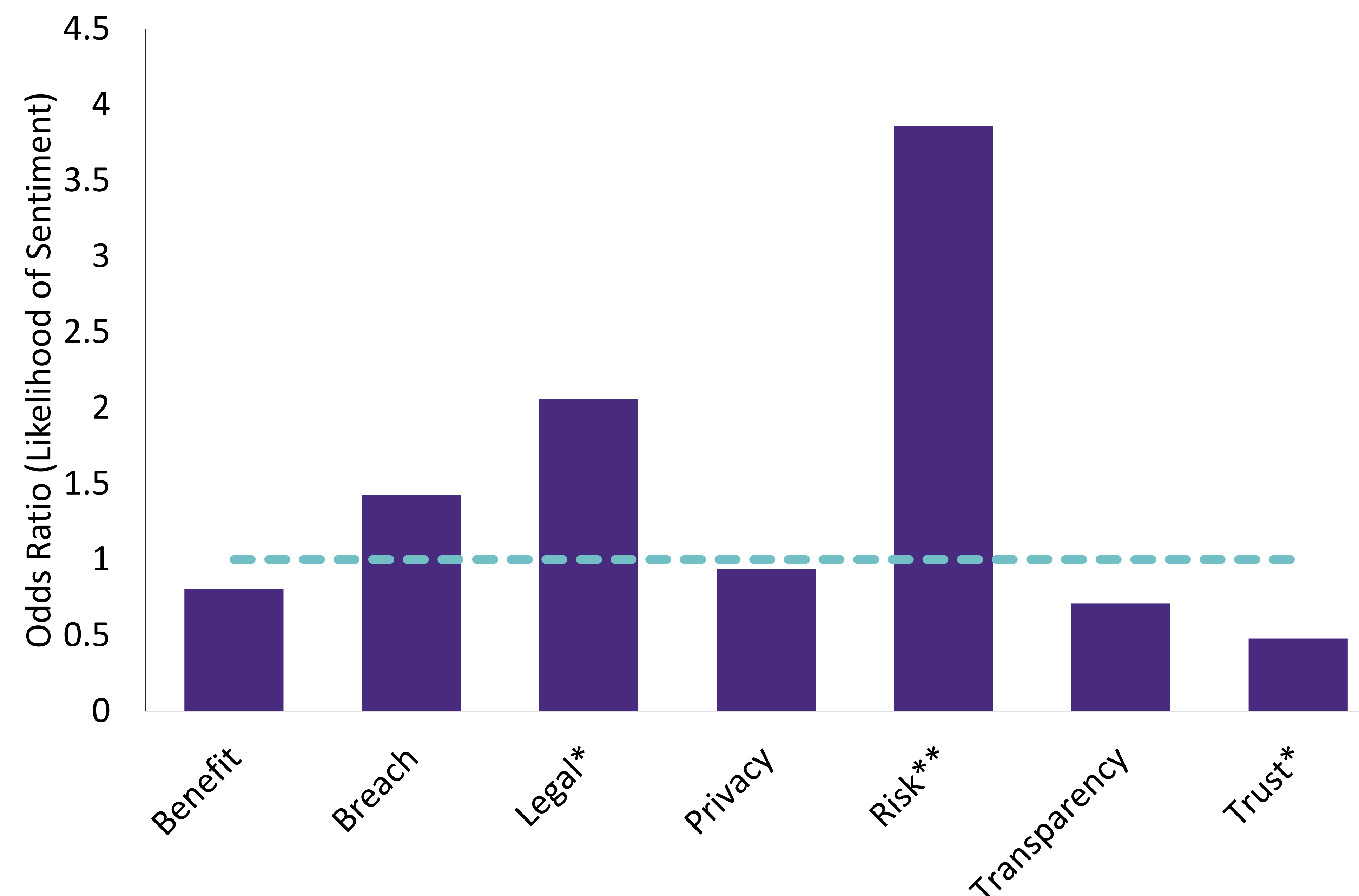
- Data were collected by employing a web scraping script that gathered 100 Reddit posts with the following keywords: ambient listening, AI scribe, privacy, trust, data, consent, transparency, risk, benefit, doctor, healthcare, clinic, clinician.
- A sentiment analysis was performed to analyze the distribution of sentiment (negative to positive).
- Then, a logistic regression was used to determine which factors were most significant in determining sentiment.

Results



Thematic Predictors of Negative Sentiment

* $p < .05$, ** $p \leq .001$



Results

- Results showed that a majority of online discourse showed neutral and positive sentiments. Trust, legal, and risk factors were found to be significant in the machine learning model with a model performance of $F1=0.72$.

Discussion

- The general public's opinions focused on concerns related to data security and transparency, contrasting with health professionals who, in previous studies, focused more on accuracy and ease of use. Additionally, the sentiment distribution was mostly uniform, showing a lack of highly polarized discourse regarding AI ambient listening.
- The current study expands the field's understanding of the underlying factors influencing patient perceptions with the use of AI and ambient listening technology. Hospitals need to address any gaps in ethics and governance to ease psychological concerns and maintain trust in providers.
- Researchers should consider different ethical frameworks and psychological constructs in future studies to improve clinical encounters for both healthcare staff and patients.
- Overall, the study demonstrates how utilizing predictive modeling and unsolicited public discourse can map real-world factors that influence the implementation of clinical tools and the dynamics of stakeholder relationships.
- Limitations of the results include the analysis emphasizing the presence rather than the intensity of concerns, multicollinearity, sampling bias, and demographics of the platform.

References & Further Information

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